

4.00 USD

1. Sentinel Smart Valve Actuator (Model S-WFC01)

- *Based on Ecowitt WFC01*
- Smart water timer valve designed to mitigate localized wildfire threat by auto-wetting property grounds when dry soil moisture thresholds trigger parametric alarm states.
- **Price:** \$79.00 USD

2. Automated Network Onboarding Workflow

The Shopify purchase acts as the initial transaction in the node onboarding lifecycle.

```
``mermaid
sequenceDiagram
    autonumber
    actor User as Policyholder / Buyer
    participant Shop as Shopify Storefront
    participant Webhook as Shopify Webhook Worker
    participant API as Chainmail API Server
    participant DB as Firestore Database
    participant Device as IoT Sentinel Device

    User->>Shop: Purchase Sentinel Kit (S-100 + S-700)
    Shop-->>User: Order Confirmation & Shipping Label Generated
    Shop->>Webhook: Event: order/paid (Payload contains email, address)
    Webhook->>API: POST /api/onboarding/provision
    activate API
```

```
API→API: Generate secure device key & pre-provision propertyId

API→DB: Write property & pre-provisioned sensors as PENDING_ACTIVATION

API-->>Webhook: Provisioning Success

deactivate API

Webhook-->>User: Onboarding Email with Device Credentials & Claim QR Code

User→Device: Mount hardware & connect to home Wi-Fi

Device→API: POST /api/ingest/ecowitt (First Telemetry Ping)

activate API

API→API: Authenticate device with secure pre-provisioned key

API→DB: Update status to VERIFIED_NODE & Activate Parametric Coverage

API-->>Device: Acknowledge (HTTP 200)

deactivate API

API→User: App Push Notification: "Node Active. Coverage Live!"

,

---
```

3. API Provisioning Endpoint Specifications

When a checkout completes, Shopify sends an order/paid webhook to our backend API server:

Endpoint: `POST /api/onboarding/provision`

Access Control: HMAC Signature verified using Shopify API Webhook Shared Secret.

Request Payload (from Shopify):

```
`json
{
  "order_id": 987654321,
```

```
"email": "customer@example.com",

"shipping_address": {

"address1": "12665 Julington Oaks Dr",

"city": "Jacksonville",

"province": "FL",

"zip": "32223",

"country": "US"

},

"line_items": [

{

"sku": "SENTINEL-KIT-A",

"quantity": 1,

"name": "Sentinel Gateway & Wittboy Array"

}

]

}
```

Provisioning Logic:

1. **Address Validation:** Call Google Address Validation API to verify CASS status and retrieve exact rooftop coordinates (lat / lng).
2. **Database Record Creation:**
 - Create a new document in /properties:

```
`typescript
```

```
{
```

```
ownerEmail: "customer@example.com",

address: "12665 Julington Oaks Dr, Jacksonville, FL 32223",

lat: 30.1421936,

lng: -81.6157784,

riskScore: 72.4, // Calculated by Underwriting Agent based on coordinates

status: "PENDING_ACTIVATION",

createdAt: serverTimestamp()

}
```

,

- Create a document in `/properties/{propertyId}/sensors/Anemometry`:

```
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{

  type: "Anemometry",

  verificationStatus: "UNVERIFIED",

  hardwareType: "Ecowitt",

  value: 0

}
```

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1. **Secret Generation:** Generate a unique password/secret token for the gateway and email it as a QR code to the user.

4. UI/UX Storefront Integration

The Shopify storefront theme will match Chainmail's core brand design system: **Cobalt & Gold / Slate & Gold**.

Theme Design System

- **Primary Background:** #030712 (Slate Black)
- **Secondary Cards:** rgba(17, 24, 39, 0.7) (Translucent Dark Blue)
- **Accents:** #D4AF37 (Metallic Gold)
- **Fonts:** Outfit (Headings) and Inter (Body)

Custom Liquid Elements

- **Network Node Counter:** A live ticker in the header showing: ACTIVE_NODES: 1,482 | TOTAL_COVERAGE_LOCKED: \$34,891,000`.
- **Map Twin Preview:** An interactive WebGL globe on the landing page showing glowing gold dots representing active coverage sensors in real time.
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The Claims Adjuster Agent: When a disaster hits, this AI looks at satellite photos to verify the damage. If it's confident, it pays the customer instantly. If it isn't sure, it sends the claim to a human to review—it is strictly programmed **never** to automatically deny a claim.

The Vigilance Oracle Agent: This AI constantly watches the weather channel. If it sees a Category 5 hurricane coming, it flags the map and readies our bank accounts to pay people before they even ask.

The Quant Brain Actuary Agent: This AI watches our bank account. If we take on too much risk in one state, it automatically buys "reinsurance" (backup insurance for us) to keep the company financially safe.

Your Day-to-Day Responsibilities

As a Product Owner, you will:

- **Monitor the Vision:** Look at the Enterprise Dashboard. Are the AI agents making decisions that align with the BallPark Initiative?
- **Prioritize the Backlog:** Work with the Product Manager to decide which neighborhood of our "city" needs improvement next.
- **Understand the Costs:** The Enterprise dashboard shows you exactly how many fractions of a cent the AI is spending to make decisions. You balance the cost of the AI against the value it provides.

You don't need to know how to code the AI. You just need to know what the AI **should** be doing to help our customers. Welcome to the team! ,500 payout cap) just so we can get their sensor data and help protect the community.

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Here is how you handle it:

- Explain that because their house is in an **Extreme Risk Zone** (like right on the beach), our Underwriting AI cannot offer them half a million dollars right now.
- However, explain that we *did not reject them*. We offered them the **IoT Baseline Policy**. This gives them ,500 of emergency cash, and in return, they install a weather sensor on their house. This helps us gather data to protect the rest of their neighborhood!

Need Help?

If the screen turns white, or the map stops loading, do not panic! Just take a screenshot and send it to your Product Manager with the Jira ticket label UI-BUG.

Welcome to Chainmail! We are glad you're here.

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2. Critical Apple App Store Requirements (iOS)

Apple is notoriously strict regarding financial and insurance applications. To pass the App Review Board, we must guarantee the following:

WARNING

digital payment & Financial Regulations (Guideline 3.1.5)

Because Chainmail utilizes automated execution rules for payouts using instant fiat rails (Visa Direct, RTP, FedNow), the app *must* be submitted by a legal entity (not an individual developer) that is registered as a financial institution or has the appropriate licenses in the regions it operates.

- **Action Item:** Ensure our corporate D-U-N-S number is used for the Apple Developer Account, and our legal entity name exactly matches the app publisher name.

IMPORTANT

Data Privacy & Account Deletion (Guideline 5.1.1)

Apple absolutely mandates that any app allowing account creation *must* also provide an in-app option to completely delete the account and all associated data.

- **Action Item:** Build an automated "Delete Account" API route in the Vercel backend.

- **App Tracking Transparency (ATT):** If we use analytics that track across third-party apps, we must implement the ATT prompt.
 - **Minimum Functionality:** The app cannot just be a "wrapped website" (Guideline 4.2). It must utilize native iOS features (e.g., Push Notifications, FaceID/TouchID for secure login, local caching).
-

3. Critical Google Play Requirements (Android)

Google Play has recently tightened restrictions on testing and financial apps.

WARNING

20-Tester Rule

New Google Play developer accounts must run a closed test with a minimum of 20 opted-in testers for at least 14 continuous days before you can apply for production access.

- **Action Item:** We must recruit 20 beta testers internally or via the Vigilance Grid network well before launch.

IMPORTANT

Financial Features Policy

Google requires a prominent "Financial Features Declaration" during submission. If the app facilitates insurance or automated payouts, we must provide links to our regulatory compliance documentation and operating licenses in our Google Play Console declaration.

- **Data Safety Section:** We must accurately declare exactly what data is collected (e.g., location data from Vigilance Grid IoT integration, financial info) and whether it is encrypted in transit.

4. Next Steps for the Development Team

To transition our current Vercel web app into a compliant Native App:

1. **Architecture:** We must decide whether to wrap the Vercel app using a framework like Capacitor/Ionic, or rebuild the frontend natively using React Native.
2. **Privacy Policy URL:** A live, hosted Privacy Policy must be created on the public-facing website.
3. **Support URL:** A dedicated support and contact page must be established for user inquiries.

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0/mo) post-funding for team access. **Database & Scaling** GCP Firestore / Blaze Plan 00.00 Scales dynamically; covers production compute and database queries. **Object Storage (Media/PII)** Cloudflare R2 \$0.00 Selected specifically to eliminate AWS/GCP data egress fees. **Geospatial & Convective Mapping** XWeather API & Vexcel Imagery \$750.00 Commercial subscription for severe convective storm mapping and high-res post-storm imagery. **Core AI Routing & HITL Verification** Google Gemini 1.5 Flash & 3.1 Pro \$65.00 Production Gemini API tokens for IoT routing and human-in-the-loop validation. **Claims Fraud Audit** ISO Verisk ClaimSearch 50.00 Active double-claim checking API subscription. **Instant Fiat Payment Rails** Stripe Visa Direct / RTP / FedNow 5.00 Transaction fees for instant payout routing. **Version Control** GitHub (Free Tier) \$0.00 Upgrade to GitHub Team (\$4/user) when engineering team expands. **Executive Compensation** COO Phase 1 (1099) \$0.00 (Equity/\$RMOR only) Transition to Full-Time Phase 1 (Salary + 8% Match) post-funding. **Total Monthly Burn** ,090.00

Future Capital Expenditure (Post-Funding)

Once institutional seed funding is secured, the following immediate upgrades will be unlocked to mature the ecosystem:

1. **CFAN Proprietary Licensing:** We will deprecate our reliance on open-source NWS/NCEP data and execute a commercial licensing agreement with the **Climate Forecast Applications Network (CFAN)**. This will drop our 5-day track error to ~131 miles, providing institutional credibility to reinsurers. *(Estimated: Negotiated Enterprise Contract)*
2. **Amazon IP Accelerator:** We will abandon the standard USPTO pending application process and utilize Amazon's vetted legal firms to fast-track our Sentinel IoT hardware Trademark and Brand Registry. *(Estimated: ,500 - \$3,000)*
3. **Auditor Integration Fees:** Funding integrations with legacy Third-Party Vendors (TPVs) to establish automated verification APIs.

0/mo) post-funding for team access. **Database & Scaling** GCP Firestore / Blaze Plan 00.00 Scales dynamically; covers production compute and database queries. **Object Storage (Media/PII)** Cloudflare R2 \$0.00 Selected specifically to eliminate AWS/GCP data egress fees. **Geospatial &**

Convective Mapping XWeather API & Vexcel Imagery \$750.00 Commercial subscription for severe convective storm mapping and high-res post-storm imagery. **Core AI Routing & HITL Verification** Google Gemini 1.5 Flash & 3.1 Pro \$65.00 Production Gemini API tokens for IoT routing and human-in-the-loop validation. **Claims Fraud Audit** ISO Verisk ClaimSearch 50.00 Active double-claim checking API subscription. **Secure Registry & Smart Contract execution** Secure L2 Ledger Mainnet 5.00 Sponsored gas for transaction settlement. **Version Control** GitHub (Free Tier) \$0.00 Upgrade to GitHub Team (\$4/user) when engineering team expands. **Executive Compensation** COO Phase 1 (1099) \$0.00 (Equity/\$RMOR only) Transition to Full-Time Phase 1 (Salary + 8% Match) post-funding. **Total Monthly Burn ,090.00**

Future Capital Expenditure (Post-Funding)

Once institutional seed funding is secured, the following immediate upgrades will be unlocked to mature the ecosystem:

- 1. **CFAN Proprietary Licensing:** We will deprecate our reliance on open-source NWS/NCEP data and execute a commercial licensing agreement with the **Climate Forecast Applications Network (CFAN)**. This will drop our 5-day track error to ~131 miles, providing institutional credibility to reinsurers. *(Estimated: Negotiated Enterprise Contract)*
- 2. **Amazon IP Accelerator:** We will abandon the standard USPTO pending application process and utilize Amazon's vetted legal firms to fast-track our Sentinel IoT hardware Trademark and Brand Registry. *(Estimated: ,500 - \$3,000)*
- 3. **DVN Node Infrastructure:** Funding the initial liquidity pools on Secure L2 Ledger to incentivize legacy Third-Party Vendors (TPVs) to transition into distributed Validator Network (DVN) node operators.

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FILING TYPE: Form, Rate, and Market Conduct Authorization (Parametric Catastrophe & TradFi Ingestion) **SUBMITTED TO:** Florida Office of Insurance Regulation (OIR) / NAIC Digital Assets Review Working Group **REGISTRANT:** Chainmail Ecosystem Protocol (as a Risk Financing Entity) **DATE OF FILING:** June 2, 2026 **DOCUMENT STATUS:** Final Submission Draft

Section I: Executive Summary & System Taxonomy

This memorandum establishes the actuarial framework, pricing algorithms, and algorithmic risk mitigation controls for the **Chainmail Ecosystem Protocol**. Chainmail operates as an automated parametric risk-transfer system designed to cover physical property and concurrent network risks without traditional human adjusters.

By utilizing real-time Internet of Things (IoT) sensors, Evolution telemetry, and public meteorological data, Chainmail calculates catastrophe (CAT) risk, pools capital reserves, and automates claims payouts under five minutes from trigger events.

1. Covered Hazards

Chainmail uses a multi-hazard parametric underwriting matrix:

- **Physical Perils:** High-velocity wind events (Anemometry), storm surge/flooding (Hydrology), and barometric pressure anomalies (Barometric) mapped to specific property boundaries.
- **Technical Perils:** Ingestion API exploits (\$P_e\$), Gateway transaction failure or telemetry corruption, and processing network latency.

2. Level of Implementation

Risk pooling and settlement are automated at the database level, backed by secure commercial bank cash reserves. Collateral is denominated in USD cash deposits, managed via automated secure escrow accounts.

Section II: Actuarial Data & Pricing Modeling

Traditional property insurance pricing relies on decades of historical claims databases (e.g., ISO Verisk). Because Chainmail covers automated parametric risks and real-time physical telemetry, the protocol establishes a dynamic actuarial table based on continuous software audit cadences and physical sensor grids.

1. Actuarial Variables & Live Metrics

The table below represents the exact structural state of Chainmail’s metrics as formatted for regulatory evaluation:

Actuarial Variable	Metric Definition	Chainmail Current Live Data	Regulatory Purpose (FL OIR Compliance)
Total Value Locked (TVL)	Total Capital Base At Risk	\$42,500,000 USD	Defines the maximum exposure limits and capital reserves for solvency testing.
RTP Routing Correlation (\$\alpha_i\$)	Interdependence of payment rails across banks	$\alpha = 0.12$ (Low Systemic Rail Outage)	Proves to the DOI that a vulnerability or exploit on Bank A will not trigger a correlated liquidity drain on Bank B.
Audit Cadence / Re-evaluation Speed	Frequency of code integrity reviews and risk profile modifications	Continuous (Evaluated every 24 hours)	Fulfills "Dynamic Risk Monitoring" requirements to prevent stale risk exposure.
Telemetry Data Streams	Automated physical and technical metrics mapped to pricing models	Real-time Call Detail Records (CDRs) / API load optimization trends	Establishes objective, untampered historical logs to baseline parametric trigger pricing.
Wind Speed Trigger Threshold	Parametric boundary for wind-related payouts	Property-specific (e.g., 90 mph constant)	Acts as the objective, non-adjustable claims trigger, bypassing human adjuster bias.

2. The Pricing Formula

The premium charged to policyholders (P) is a function of the Parametric Event Probability (P_{cat}), the API Service Exploit Probability (P_e), the RTP Routing Correlation (α), and an administrative loading factor (λ):

$$P = \left(P_{cat} \times \text{Coverage Limit} \right) + \left(P_e \times \text{TVL} \times (1 + \alpha) \right) + \lambda$$

Section III: Algorithmic Verification & NAIC Compliance

To address the "black-box" underwriting problem, state regulators enforce transparency principles regarding automated models. Chainmail aligns with the National Association of Insurance Commissioners (NAIC) principles for Artificial Intelligence Systems (AIS).

1. Explainability and Transparency

Chainmail's underwriting algorithms provide a clear, step-by-step audit trail for every quote. If a property's annual premium is adjusted, the API output matches the specific variables:

- Local 50-year storm recurrence interval (NOAA GRIB2 grids).
- Structural profiling data (roof age, construction type).
- Live local IoT sensor health status.

Regulators can verify these parameters at any time via automated compliance logs, eliminating hidden model assumptions.

2. Algorithmic Bias and Discrimination Mitigation

Our algorithms utilize strictly non-discriminatory inputs (geospatial coordinates, sensor metrics, and code statistics). Underwriting grades (\$A+, A, B\$) are determined using physical risk assessments (soil liquefaction, elevation, wind resistance) and do not incorporate demographic, socioeconomic, or zip-code proxies.

3. Human-in-the-Loop (HITL) Actuarial Oversight

Although underwriting and settlement are automated, Chainmail implements a hybrid governance model:

- **Appointed Actuary:** A qualified Fellow of the Casualty Actuarial Society (FCAS) reviews the protocol's pricing curves quarterly.
- **Emergency Circuit Breaker:** If the sentiment analysis model detects systemic contract anomalies, or if a physical oracle failure is detected, the Appointed Actuary can manually halt the automated quote system and trigger a manual review.

Section IV: Asset Adequacy & Stress Testing (AG 53 Compliance)

Under **NAIC Actuarial Guideline 53 (AG 53)**, insurance entities holding complex collateral must demonstrate capital sufficiency under stress conditions.

1. Sensitivity Analysis Parameters

Chainmail conducts automated stress-testing simulations daily across three severe scenarios:

- **Systemic Liquidation Event:** A simulated **75% drop** in secondary reserve yields within 24 hours. The model demonstrates that the \$42,500,000 USD core pool remains fully liquid and isolated from secondary market volatility due to reserve ratio requirements.

- **Decoupling Event:** A simulated delay in a specific RTP payment rail. The protocol's ingestion engine automatically routes settlement reserves to alternative payment rails (e.g., FedNow/Stripe Visa Direct) to guarantee payout values.
- **Conforming Peril Clustered CAT (Florida Hurricane Strike):** A simulated Level 4 hurricane striking multiple insured properties simultaneously. Since our RTP Routing Correlation coefficient is low ($\alpha = 0.12$), the capital backstop pools remain solvent even if 45% of active parametric contracts trigger payouts within the same 48-hour window.

2. Reserves Adequacy Certification

Our Appointed Actuary issues a quarterly *Statement of Actuarial Opinion (SAO)* verifying that Chainmail's reserves exceed the 99.5% Value-at-Risk (VaR) threshold over a 1-year horizon.

Section V: The Three-Tier Data Taxonomy

Chainmail systematically categorizes all incoming data into a structured taxonomy. This allows the risk team to predict technical anomalies before a systemic contract exploit occurs.

```
graph TD
    Data[Chainmail Data Taxonomy] --> Exhaust[Exhaust Data]
    Data --> Digital[Digital Content Data]
    Data --> Sensing[Sensing Data]
    Exhaust --> E1[API Response Time Trends]
    Exhaust --> E2[Transaction Latencies]
    Exhaust --> E3[Automated Payout Execution Volumes]
    Digital --> D1[Compliance API Registry History]
    Digital --> D2[GitHub Commits / Code Modifications]
    Digital --> D3[Audit Trail Registry Logs]
    Sensing --> S1[IoT Wind Sensors]
    Sensing --> S2[Barometric Oracles]
    Sensing --> S3[Plaid Proof of Reserve Integration]
```

1. Exhaust Data (Passive Telemetry)

- **Definition:** Passive metadata generated through API requests and computing resources.
- **Actuarial Use:** Spikes in execution latencies or API error rates are analyzed in real-time to flag possible DDoS or injection attacks before they compromise the system.

2. Digital Content Data (Ledger & Code History)

- **Definition:** Public database transaction history, audit logs, and code commits.
- **Actuarial Use:** Tracks compiler versions, library dependencies, and security patches. Each code commit triggers an automated dry-run validation of the pricing rules.

3. Sensing Data (Real-time Telemetry)

- **Definition:** Active inputs fetched from the physical IoT sensor network and automated weather databases.
- **Actuarial Use:** Determines whether the trigger wind speed or barometric threshold has been breached at a specific property location.

Section VI: Commercial Protection & Replicability

Regulators require assurance that Chainmail's algorithms are proprietary, commercially protected, and secure from external manipulation.

1. Proprietary Boundaries

- **Open-Source Layer:** The parametric settlement rules, API integration code, and security access rules are fully open-source on GitHub to ensure complete trust, transparency, and public auditability.
- **Proprietary IP Layer:** The internal risk scoring engines, historical CAT regression models, and sentiment-based fraud-detection nodes remain closed-source and are secured within corporate repositories.

2. Replicability & Copycat Defenses

To prevent adversarial copycats from duplicating our actuarial calculations to drain pool liquidity:

- Every parametric contract is cryptographically signed using a unique physical hardware security module (HSM) oracle key.
- Pricing updates require authorized multi-party validation state adjustments, ensuring that a duplicated system cannot replicate our settlement triggers without our validated data feeds.

Section VII: Regulatory Compliance Checklist

Required Asset	Regulatory Purpose	Status / Provider
Statement of Actuarial Opinion (SAO)	Certifies pool reserves are sufficient to cover maximum potential losses.	Appointed Actuary (FCAS certified)
Certified API Code Audits	Validates code security against malicious exploits.	Independent Cybersecurity Firm
AG 53 Asset Suitability Analysis	Proves the USD reserves can survive a severe market crash.	Quantitative Risk Advisory Team
AIS Governance Program Documentation	Proves automated underwriting rules comply with NAIC explainability principles.	Chainmail Legal & Compliance Team